Study Design, Objectives and Endpoints

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Study Design:

A5346 is a phase II, randomized, double-blinded, placebo-controlled trial of a DPP4 inhibitor (sitagliptin; 100 mg/day) to determine whether sitagliptin reduces plasma sCD14 concentrations.

The study population will include HIV-infected men and women >=18 years of age who are continuously treated with cART and have undetectable plasma viremia for the prior >= 48 weeks.

Participants will be stratified by screening CD4+T cell count and statin use and randomized 1:1 to 2 arms such that 43 participants will be enrolled in each arm.

Arm 1: Sitagliptin (Januvia) 100 mg one tablet daily p.o. for 16 weeks, followed by a 4-week post-treatment follow-up.

Arm 2: Placebo for sitagliptin one tablet daily p.o. for 16 weeks, followed by a 4-week post-treatment follow-up.

(Section 3.0)

Study Objectives:

Primary: To assess the effect of 16 weeks of sitagliptin 100 mg/day on plasma sCD14 concentrations.

(Section 1.2)

Secondary:

1. To assess the effect of 16 weeks of sitagliptin 100 mg/day on plasma concentrations of CXCL10 (IP-10), sCD163, IL-6, sTNFR1 & 2, hs-CRP, and sCD26.

2. To assess the effect of 16 weeks of sitagliptin 100 mg/day on the percentages of CD4+CD38+HLA-DR+ and CD8+CD38+HLA-DR+ T-cell subpopulations.

3. To assess the effect of 16 weeks of sitagliptin 100 mg/day on the percentages of CD14+CD16- (classical), CD14+CD16+ (intermediate), and CD14dimCD16++ (non-classical) monocyte subpopulations.

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4. To assess the effect of 16 weeks of sitagliptin 100 mg/day on the CD4+/CD8+ T-cell ratio.

5. To assess the safety of 16 weeks of sitagliptin 100 mg/day in HIV-infected persons on ART.

6. To assess the durability of the effect of sitagliptin 100 mg/day on plasma sCD14 concentrations 4 weeks after discontinuing sitagliptin or placebo (week 20).

7. To assess the durability of the effect of sitagliptin 100 mg/day on the percentages of CD4+CD38+HLA-DR+ and CD8+CD38+HLA-DR+ T-cell subpopulations 4 weeks after discontinuing sitagliptin or placebo (week 20).

8. To assess the durability of the effect of sitagliptin 100 mg/day on the percentages of CD14+CD16- (classical), CD14+CD16+ (intermediate), CD14dimCD16++ (non-classical) monocyte subpopulations 4 weeks after discontinuing sitagliptin or placebo (week 20).

9. To assess the durability of the effect of sitagliptin 100 mg/day on the CD4+/CD8+ T-cell ratio 4 weeks after discontinuing sitagliptin or placebo (week 20).

10. In the event that sitagliptin reduces plasma markers of immune activation or inflammation, we will assess the effect and durability of sitagliptin 100 mg/day on the transcriptional activation of cell-associated HIV-1, the PBMC HIV-1 RNA/DNA ratio.

11. In the event that sitagliptin reduces plasma markers of immune activation or inflammation, we will assess the effect and durability of sitagliptin 100 mg/day on monocyte mRNA expression for pro-inflammatory cytokine and chemokine markers (e.g., TLR-2, TLR-4, CCR2, and TNFalpha).

(Section 1.3)

Study Endpoints:

Primary, Immunogenicity: Change in sCD14 from baseline to week 16.

(Section 9.2.1)

Secondary, Immunogenicity: 1 Change from baseline to week 16 in circulating levels of inflammatory
biomarkers CXCL10 (IP-10), sCD163, IL-6, sTNFR1 & 2, hs-CRP, sCD26.

2. Change from baseline to week 16 in the percentage

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and number of
CD4+CD38+HLA-DR+ and CD8+CD38+HLA-DR+ T-cell subpopulations

3. Change from baseline to week 16 in the percentages and number of
CD14+CD16- (classical), CD14+CD16+ (intermediate) and CD14dim
CD16++ (non-classical) monocyte subpopulations.

4. Change from baseline to week 16 in CD4+/CD8+ T-cell ratio.

5. Occurrence of Grade >=3 AEs and selected Grade >=2 AEs related to
study drug.

6. Change from week 16 to week 20 for: plasma sCD14 levels, plasma
biomarker levels (CXCL10 (IP-10), sCD163, IL-6, sTNFR1 & 2, hs-CRP,
sCD26), percentage and number of T-cell and monocyte subpopulations,
and CD4+/CD8+ T-cell ratio.

7. In the event that sitagliptin reduces plasma markers of immune activation
or inflammation, change from baseline to week 16 and week 16 to week 20
in monocyte mRNA expression for pro-inflammatory cytokine or chemokine
markers (e.g., TLR-2, TLR-4, CCR2, TNFalpha).

8. In the event that sitagliptin reduces plasma markers of immune activation
or inflammation, change from baseline to week 16 and week 16 to week 20
in monocyte HIV RNA/DNA ratio.

(Section 9.2.2)

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